

Image Processing and Pattern Recognition

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Lecture – 07

Harimohan Khatri

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In this lecture...

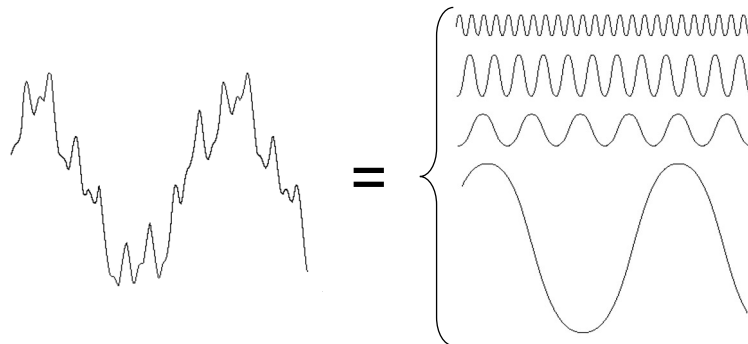
Enhancement in Frequency domain

- i. Introduction to Fourier Transform, DFT, FFT
- ii. Computing and visualizing 2D DFT

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Fourier Series

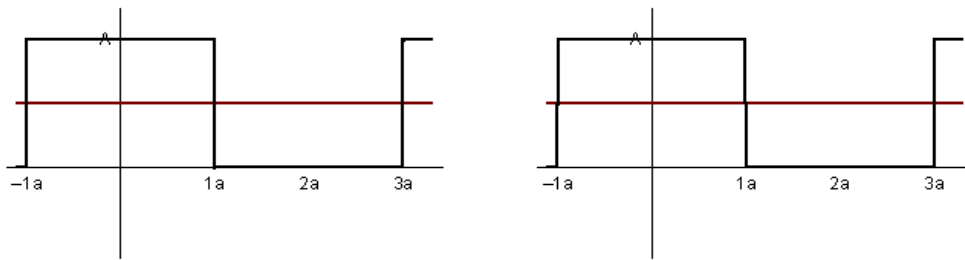
- Fourier Series: Any function that periodically repeats itself can be expressed as a sum of sines and cosines of different frequencies each multiplied by a different coefficient.



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Fourier Series

Closer is the original function as we add more and more frequencies



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Fourier Transform

- The Fourier transform of a continuous function $f(t)$ of a continuous variable t , denoted $\mathcal{F}\{f(t)\}$ is defined by the equation

$$F\{f(x)\} = F(u) = \int_{-\infty}^{\infty} f(t) e^{-j2\pi ux} dx$$

- Given the $F(u)$, $f(x)$ can be obtained by using inverse Fourier transform:

$$\begin{aligned} \mathcal{F}^{-1}\{F(u)\} &= f(x) \\ &= \int_{-\infty}^{\infty} F(u) e^{j2\pi ux} du \end{aligned}$$

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Fourier Transform

- Fourier transform of real function is generally complex. So,

$$F(u) = R(u) + j I(u)$$

where $R(u)$ and $I(u)$ denote the real and imaginary components of $F(u)$ respectively.

- Expressed in exponential form, $F(u)$ is:

$$F(u) = |F(u)| e^{j\varphi(u)}$$

where $|F(u)| = \sqrt{R^2(u) + I^2(u)}$ and $\varphi(u) = \tan^{-1} \left[\frac{I(u)}{R(u)} \right]$

- The magnitude function $|F(u)|$ is called Fourier spectrum of $f(x)$ and $\varphi(u)$ is the phase angle.

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Fourier Transform

- The square of the spectrum,

$$P(u) = |F(u)|^2 = R^2(u) + I^2(u)$$

is commonly called the power spectrum(or spatial density) of $f(x)$.

- The variable u is often called the frequency variable. This name arises from the expression of the exponential term $e^{-j2\pi ux}$ in terms of sines and cosines (From Euler's Formula):

$$e^{-j2\pi ux} = \cos(2\pi ux) - j \sin(2\pi ux)$$

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Fourier Transform

- Interpreting the integral in the Fourier Transform equation as a limit summation of discrete terms;
 - $F(u)$ is composed of an infinite sum of sine and cosine terms
 - Each value of u determines the frequency of its corresponding sine-cosine pair.
- Considering a step function as;

$$f(x) = \begin{cases} A & \text{for } 0 \leq x \leq X \\ 0 & \text{else} \end{cases}$$

$$\begin{aligned} F(u) &= \int_{-\infty}^{\infty} f(x) e^{-j2\pi ux} dx \\ &= \int_0^X A e^{-j2\pi ux} dx \\ &= \frac{-A}{j2\pi u} [e^{-j2\pi ux}]_0^X \\ &= \frac{-A}{j2\pi u} [e^{-j2\pi uX} - 1] \\ &= \frac{A}{j2\pi u} [e^{j\pi uX} - e^{-j\pi uX}] e^{-j\pi uX} \\ &= \frac{A}{\pi u} \sin(\pi uX) e^{-j\pi uX} \end{aligned}$$

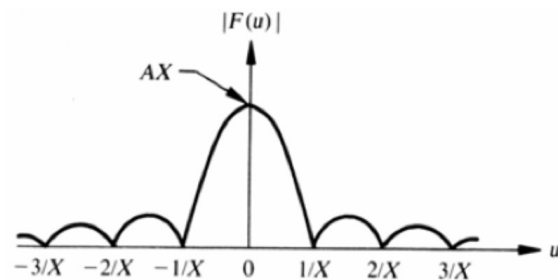
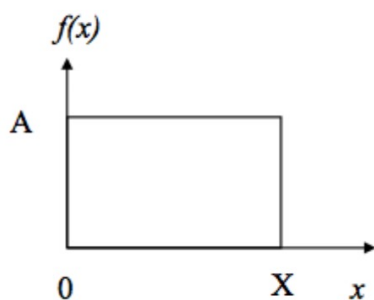
The Fourier spectrum is:

$$\begin{aligned} |F(u)| &= \left| \frac{A}{\pi u} \sin(\pi uX) \right| e^{-j\pi uX} \\ &= AX \left| \frac{\sin(\pi uX)}{\pi uX} \right| \end{aligned}$$

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Fourier Transform

- The plot of step function and its Fourier transform spectrum is as:



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2-D Fourier Transform

- The Fourier transform can be extended to two dimensions:

$$\mathcal{F}\{f(x,y)\} = F(u,v) = \iint_{-\infty}^{\infty} f(x,y) e^{-j2\pi(ux+vy)} dx dy$$

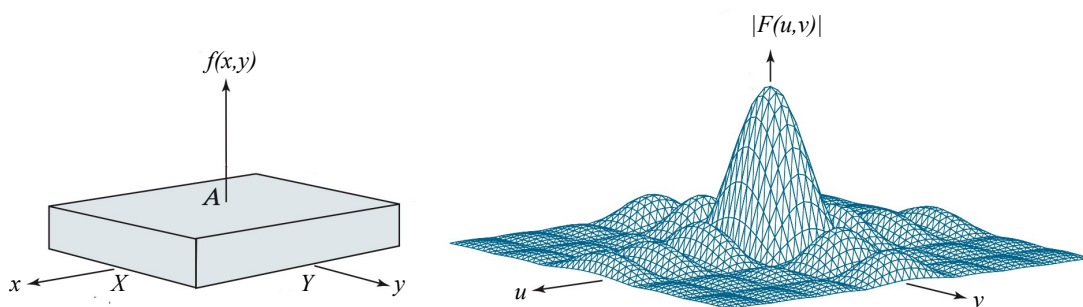
and inverse transform

$$\mathcal{F}^{-1}\{F(u,v)\} = f(x,y) = \iint_{-\infty}^{\infty} F(u,v) e^{j2\pi(ux+vy)} du dv$$

- 2-D Fourier Spectrum is: $|F(u,v)| = \sqrt{R^2(u,v) + I^2(u,v)}$
- The phase angle is $\varphi(u,v) = \tan^{-1} \left[\frac{I(u,v)}{R(u,v)} \right]$
- The power spectrum is: $P(u,v) = |F(u,v)|^2 = R^2(u,v) + I^2(u,v)$

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2-D Fourier Transform



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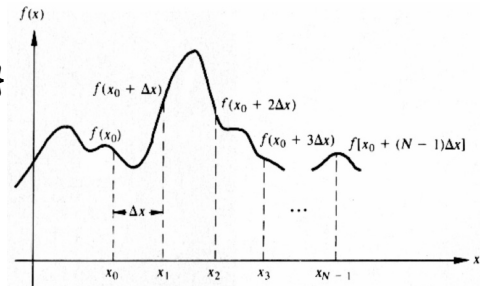
Discrete Fourier Transform

- Suppose a continuous function, $f(x)$, is discretized into a sequence, $\{f(x_0), f(x_0+\Delta x), f(x_0+2\Delta x), \dots, f(x_0+[N-1]\Delta x)\}$ by taking N samples Δx units apart.

- Given the function,

$$f(x) = f(x_0 + x \Delta x)$$

where x assumes the discrete values $0, 1, \dots, N-1$ and $\{f(0), f(1), \dots, f(N-1)\}$ denotes any N uniformly spaced samples from corresponding continuous function.



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Discrete Fourier Transform

- Discrete fourier transform is given by:

$$F(u) = \frac{1}{N} \sum_{x=0}^{N-1} f(x) e^{\frac{-j2\pi ux}{N}} \text{ for } u = 0, 1, \dots, N-1$$

- The discrete Inverse Fourier transform is given by:

$$F(x) = \sum_{u=0}^{N-1} F(u) e^{\frac{j2\pi ux}{N}} \text{ for } x = 0, 1, \dots, N-1$$

- The values of $u=0, 1, \dots, N-1$ in the discrete case correspond to samples of the continuous transform at $0, \Delta u, 2\Delta u, \dots, (N-1)\Delta u$.

- Δu and Δx are related by $\Delta u = \frac{1}{N\Delta x}$

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2-D Discrete Fourier Transform

- In the 2-D case:

$$F(u,v) = \frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x,y) e^{-j2\pi(\frac{ux}{M} + \frac{vy}{N})}$$

for $u = 0 \rightarrow M-1$ and $v = 0 \rightarrow N-1$

- The discrete Inverse Fourier transform is given by:

$$f(x,y) = \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u,v) e^{j2\pi(\frac{ux}{M} + \frac{vy}{N})}$$

for $x = 0 \rightarrow M-1$ and $y = 0 \rightarrow N-1$

- The discrete function $f(x,y)$ represents samples of the continuous samples of the continuous function at $f(x_0+x\Delta x, y_0+y\Delta y)$
- $\Delta u = \frac{1}{M\Delta x}$ and $\Delta v = \frac{1}{N\Delta y}$

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2-D Discrete Fourier Transform

- For Square Image ($M=N$):

$$F(u,v) = \frac{1}{N} \sum_{x=0}^{N-1} \sum_{y=0}^{N-1} f(x,y) e^{-j2\pi(\frac{ux+vy}{N})}$$

- And Inverse Transform is:

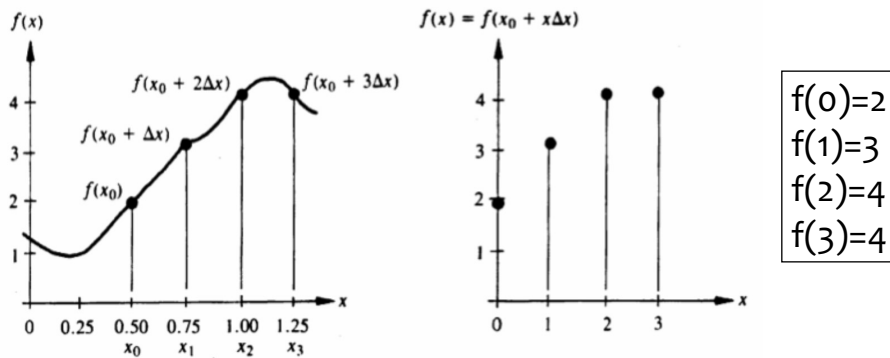
$$f(x,y) = \frac{1}{N} \sum_{u=0}^{N-1} \sum_{v=0}^{N-1} F(u,v) e^{j2\pi(\frac{ux+vy}{N})}$$

Note: $\frac{1}{N}$ symmetric normalization is used for both transform.

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2-D Discrete Fourier Transform Example

Calculate the Fourier spectrum for given function: Consider sampling at $x_0 = 0.5$, $x_1 = 0.75$, $x_2 = 1.0$ and $x_3 = 1.25$



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2-D Discrete Fourier Transform Example

$$F(u) = \frac{1}{N} \sum_{x=0}^{N-1} f(x) e^{\frac{-j2\pi ux}{N}} \text{ for } u = 0, 1, 2, 3.$$

➤ Calculating for each terms:

$$\begin{aligned} F(0) &= \frac{1}{4} \sum_{x=0}^3 f(x) e^{\frac{-j2\pi 0x}{4}} \\ &= \frac{1}{4} \sum_{x=0}^3 f(x) e^0 \\ &= \frac{1}{4} [f(0) + f(1) + f(2) + f(3)] \\ &= \frac{1}{4} [2 + 3 + 4 + 4] = 3.25 \end{aligned}$$

$$F(2) = -\frac{1}{4} [1 + 0j]$$

$$F(3) = -\frac{1}{4} [2 + j]$$

$$\begin{aligned} F(1) &= \frac{1}{4} \sum_{x=0}^3 f(x) e^{\frac{-j2\pi 1x}{4}} \\ &= \frac{1}{4} [f(0) e^{\frac{-j2\pi 0}{4}} + f(1) e^{\frac{-j2\pi 1}{4}} \\ &\quad + f(2) e^{\frac{-j2\pi 2}{4}} + f(3) e^{\frac{-j2\pi 3}{4}}] \\ &= \frac{1}{4} [2e^0 + 3e^{\frac{-j\pi}{2}} + 4e^{-j\pi} + 4e^{\frac{-j3\pi}{2}}] \\ &= \frac{1}{4} [-2 + j] \end{aligned}$$

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2-D Discrete Fourier Transform Example

➤ Calculating Fourier Spectrum:

$$|F(u)| = \sqrt{R^2(u) + I^2(u)}$$

$$|F(0)| = 3.25$$

$$|F(1)| = \sqrt{\left(-\frac{2}{4}\right)^2 + \left(\frac{1}{4}\right)^2} = \frac{\sqrt{5}}{4}$$

$$|F(2)| = \sqrt{\left(\frac{1}{4}\right)^2 + \left(\frac{0}{4}\right)^2} = \frac{1}{4}$$

$$|F(3)| = \sqrt{\left(\frac{2}{4}\right)^2 + \left(\frac{1}{4}\right)^2} = \frac{\sqrt{5}}{4}$$

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2-D Discrete Fourier Transform

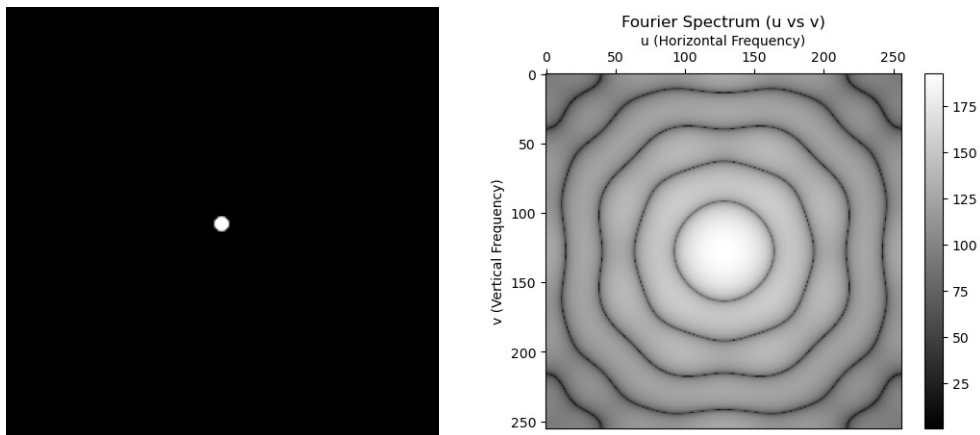
- A common technique to display Fourier Spectrum is:

$$D(u,v) = c \log[1 + |F(u,v)|]$$

where c is scaling factor and the logarithm function performs a compression of data, which is usually chosen to scale the data into the range of display device, $[0-255]$ for grayscale image.

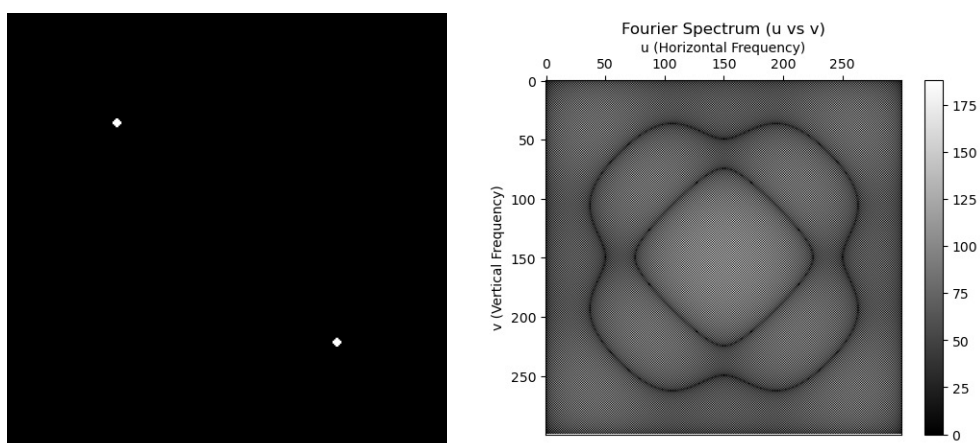
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2-D Discrete Fourier Transform Example



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2-D Discrete Fourier Transform Example



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Properties of 2-D DFT (Separability)

- We have DFT as:

$$F(u,v) = \frac{1}{N} \sum_{x=0}^{N-1} \sum_{y=0}^{N-1} f(x,y) e^{-j2\pi(\frac{ux+vy}{N})}$$

- The exponential term can be separated as:

$$e^{-j2\pi(\frac{ux+vy}{N})} = e^{\frac{-j2\pi ux}{N}} \cdot e^{\frac{-j2\pi vy}{N}}$$

- Now we can write discrete transform pair in separable forms as:

$$F(u,v) = \frac{1}{N} \sum_{x=0}^{N-1} e^{\frac{-j2\pi ux}{N}} \sum_{y=0}^{N-1} f(x,y) e^{-j2\pi(\frac{vy}{N})}$$

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Properties of 2-D DFT (Separability)

- Similarly for IDFT:

$$F(x,y) = \frac{1}{N} \sum_{u=0}^{N-1} \sum_{v=0}^{N-1} f(u,v) e^{j2\pi(\frac{ux+vy}{N})}$$

- The exponential term can be separated as:

$$e^{j2\pi(\frac{ux+vy}{N})} = e^{\frac{j2\pi ux}{N}} \cdot e^{\frac{j2\pi vy}{N}}$$

- Now we can write discrete transform pair in separable forms as:

$$F(x,y) = \frac{1}{N} \sum_{x=0}^{N-1} e^{\frac{j2\pi ux}{N}} \sum_{y=0}^{N-1} f(u,v) e^{j2\pi(\frac{vy}{N})}$$

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Properties of 2-D DFT (Separability)

- 1D-DFT over y: For each fixed x, compute a 1D FFT along the y-dimension

$$G(x,v) = \sum_{y=0}^{N-1} f(x,y) e^{-j2\pi(\frac{vy}{N})}$$

This is intermediate result $G(x,v)$ is a function of x and v.

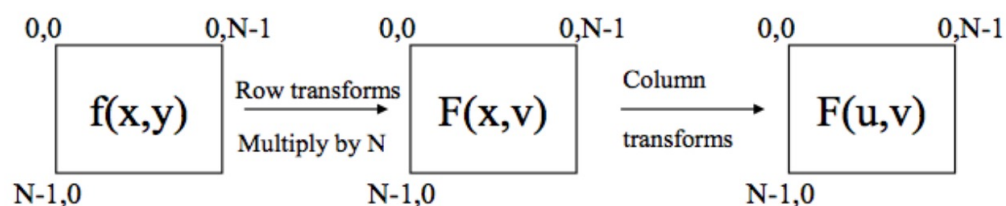
- 1D-DFT over x: for each v, compute a 1D DFT of $G(x,v)$ along the x-dimension:

$$F(u,v) = \frac{1}{N} \sum_{x=0}^{N-1} G(x,v) e^{-j2\pi(\frac{ux}{N})}$$

$\frac{1}{N}$ is normalizaiton factor for DFT

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Properties of 2-D DFT (Separability)



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Properties of 2-D DFT (Translation)

- Translation property of Fourier transform pair are:

$$f(x, y) e^{j2\pi(\frac{u_0 x + v_0 y}{N})} \Leftrightarrow F(u - u_0, v - v_0)$$

and

$$f(x - x_0, y - y_0) \Leftrightarrow F(u, v) e^{-j2\pi(\frac{u x_0 + v y_0}{N})}$$

- Multiplying $f(x, y)$ by the exponential and taking the transform results in a shift the origin of the frequency plane to the point (u_0, v_0)

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Properties of 2-D DFT (Translation)

Translation to center of the frequency rectangle:

- For this purpose, $u_0 = v_0 = N/2$;

$$e^{j2\pi(\frac{u_0 x + v_0 y}{N})} = e^{j\pi(\frac{x+y}{N})} = (-1)^{x+y}$$

and,

$$f(x, y) (-1)^{x+y} \Leftrightarrow F(u - \frac{N}{2}, v - \frac{N}{2})$$

- So, the origin of the Fourier transform of $f(x, y)$ can be moved to the center of the corresponding $N \times N$ simply by multiplying $f(x, y)$ by $(-1)^{x+y}$ before taking the transform.
- This doesnot affect the magnitude of the fourier transfrom.

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Properties of 2-D DFT (Periodicity)

- The discrete Fourier transform and its inverse are periodic with period N .

$$F(u,v) = F(u+N,v) = F(u,v+N) = F(u+N,v+N)$$
 and

$$F(x,y) = F(x+N,y) = F(x,y+N) = F(x+N,y+N)$$
- Although $F(u,v)$ repeats itself infinitely for many values of (u,v) only N values of each variable are required to obtain $f(x,y)$ from $F(u,v)$
- That means only one period of the transform is necessary to specify $F(u,v)$ in the frequency domain. Similarly for $f(x,y)$ in the spatial domain.

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Properties of 2-D DFT (Conjugate Symmetry)

- Complex conjugate of a complex number is a number with the same real part but the opposite imaginary part.
- For example, $z=a+jb$ is complex number then, $z^*=a-jb$ is the conjugate.
- If $f(x,y)$ is real, the Fourier transform exhibits conjugate symmetry

$$F(u,v) = F^*(-u,-v)$$
 or, $|F(u,v)| = |F(-u,-v)|$
 where $F^*(u,v)$ is the complex conjugate of $f(u,v)$

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Properties of 2-D DFT (Conjugate Symmetry)

	Spatial Domain [†]	Frequency Domain [‡]
1)	$f(x,y)$ real	$\Leftrightarrow F^*(u,v) = F(-u,-v)$
2)	$f(x,y)$ imaginary	$\Leftrightarrow F^*(-u,-v) = -F(u,v)$
3)	$f(x,y)$ real	$\Leftrightarrow R(u,v)$ even; $I(u,v)$ odd
4)	$f(x,y)$ imaginary	$\Leftrightarrow R(u,v)$ odd; $I(u,v)$ even
5)	$f(-x,-y)$ real	$\Leftrightarrow F^*(u,v)$ complex
6)	$f(-x,-y)$ complex	$\Leftrightarrow F(-u,-v)$ complex
7)	$f^*(x,y)$ complex	$\Leftrightarrow F^*(-u,-v)$ complex
8)	$f(x,y)$ real and even	$\Leftrightarrow F(u,v)$ real and even
9)	$f(x,y)$ real and odd	$\Leftrightarrow F(u,v)$ imaginary and odd
10)	$f(x,y)$ imaginary and even	$\Leftrightarrow F(u,v)$ imaginary and even
11)	$f(x,y)$ imaginary and odd	$\Leftrightarrow F(u,v)$ real and odd
12)	$f(x,y)$ complex and even	$\Leftrightarrow F(u,v)$ complex and even
13)	$f(x,y)$ complex and odd	$\Leftrightarrow F(u,v)$ complex and odd

Some symmetry properties of the 2-D DFT and its inverse.
 $R(u,v)$ and $I(u,v)$ are the real and imaginary parts of $F(u,v)$, respectively

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Properties of 2-D DFT (Periodicity)

- DFT of a sequence is periodic with period equal to the length of the sequence, i.e., $F[x] = F[x+N]$.

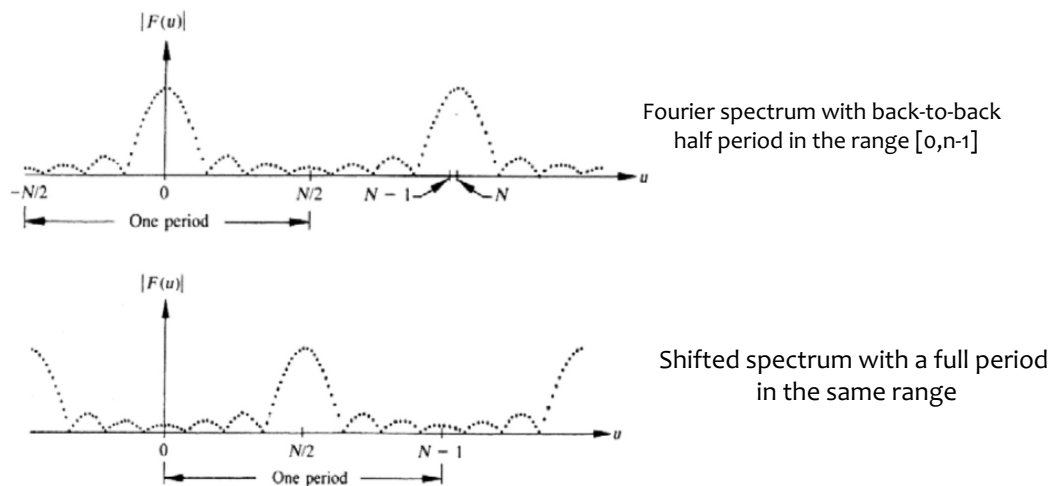
- In 2-D DFT:

$$F(u,v) = F(u+k_1M,v) = F(u,v+k_2N) = F(u+k_1M,v+k_2N)$$

$$f(x,y) = f(x+k_1M,y) = f(x,y+k_2N) = f(x+k_1M,y+k_2N)$$

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Properties of 2-D DFT (Periodicity)



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Properties of 2-D DFT (Distributivity & Scalability)

- The Fourier transform and its inverse are distributive over addition but not over multiplication. So,

$$\mathcal{F}\{f_1(x,y) + f_2(x,y)\} = \mathcal{F}\{f_1(x,y)\} + \mathcal{F}\{f_2(x,y)\}$$

$$\mathcal{F}\{f_1(x,y) \times f_2(x,y)\} \neq \mathcal{F}\{f_1(x,y)\} \times \mathcal{F}\{f_2(x,y)\}$$

- For two scalars a and b ,

$$af(x,y) \Leftrightarrow aF(u,v)$$

$$f(ax,by) \Leftrightarrow \frac{1}{|ab|} F\left(\frac{u}{a}, \frac{v}{b}\right)$$

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Properties of 2-D DFT (Average Value)

- Widely used expression for the average value of a 2-D discrete function is:

$$\bar{f}(x, y) = \frac{1}{N^2} \sum_{x=0}^{N-1} \sum_{y=0}^{N-1} f(x, y)$$

- The DC component $F(0,0)$ of the image's Fourier transform holds the sum of all pixel intensities.
- Dividing by the number of pixels gives the average value of the image. For $u=v=0$,

$$F(0,0) = \frac{1}{N} \sum_{x=0}^{N-1} \sum_{y=0}^{N-1} f(x, y)$$

Therefore, $\bar{f}(x, y) = \frac{1}{N} F(0,0)$

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Properties of 2-D DFT (Correlation)

- Correlation operation in spatial domain is:

$$(f \star h)(x, y) = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} f^*(m, n) h(x + m, y + n)$$

- Then correlation theorem states:

$$(f \star h)(x, y) \Leftrightarrow (F^* \cdot H)(u, v)$$

$$(f^* \cdot h)(x, y) \Leftrightarrow (1/MN)[(F \star H)(u, v)]$$

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Properties of 2-D DFT (Convolution)

- Convolution operation in spatial domain is:

$$(f \star h)(x, y) = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} f(m, n) h(x - m, y - n)$$

- Then convolution theorem states:

$$f \star h(x, y) \Leftrightarrow (F \bullet H)(u, v)$$

$$(f \bullet h)(x, y) \Leftrightarrow (1/MN)[(F \star H)(u, v)]$$

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Fast Fourier Transform

- Fast Fourier transform reduces the complexity of 2-D Fourier tranform $(MN)^2$ to $(MN)\log_2(MN)$ multiplications and additions.
- The fourier transform can be expressed as:

$$F(u) = \sum_{x=0}^{M-1} f(x) e^{-j2\pi ux/M} \quad u = 0, 1, 2, \dots, M-1 \quad (1)$$

- Given equation can be expressed in form of:

$$F(u) = \sum_{x=0}^{M-1} f(x) W_M^{ux} \quad \text{for } u = 0, 1, 2, \dots, M-1 \quad (2)$$

$$\text{where } W_M = e^{-j2\pi/M} \quad (3)$$

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Fast Fourier Transform

- and M is assumed to be of the form

$$M = 2^p \quad (4)$$

where p is a positive integer.

- Then it follows that M can be expressed as

$$M = 2^K \quad (5)$$

with K being a positive integer

- From (2) and (5)

$$F(u) = \sum_{x=0}^{2K-1} f(x) W_{2K}^{ux} = \sum_{x=0}^{K-1} f(2x) W_{2K}^{u(2x)} + \sum_{x=0}^{K-1} f(2x+1) W_{2K}^{u(2x+1)} \quad (6)$$

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Fast Fourier Transform

- From (3) $W_{2K}^{2ux} = W_K^{ux}$, so equation (6) can be written as,

$$F(u) = \sum_{x=0}^{K-1} f(2x) W_K^{ux} + \sum_{x=0}^{K-1} f(2x+1) W_K^{ux} W_{2K}^u \quad (7)$$

- Defining even and odd functions,

$$F_{\text{even}}(u) = \sum_{x=0}^{K-1} f(2x) W_K^{ux} \quad \text{for } u = 0, 1, 2, \dots, K-1 \text{ and}$$

$$F_{\text{odd}}(u) = \sum_{x=0}^{K-1} f(2x+1) W_K^{ux} \quad (8)$$

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Fast Fourier Transform

- For $u = 0, 1, 2, \dots, K-1$ reduces equation (7) to

$$F(u) = F_{\text{even}}(u) + F_{\text{odd}}(u)W_{2K}^u \quad (9)$$

- Because $W_M^{u+K} = W_K^u$ and $W_{2K}^{u+K} = -W_{2K}^u$ it follows,

$$F(u + K) = F_{\text{even}}(u) - F_{\text{odd}}(u)W_{2K}^u \quad (10)$$

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Fast Fourier Transform

- An M-point DFT can be computed by dividing the original expression into two parts, as indicated in (9) and (10).
- Computing the first half of $F(u)$ requires evaluation of the two $(M/2)$ -point transforms given in Eqs. (8).
- The resulting values of $F_{\text{even}}(u)$ and $F_{\text{odd}}(u)$ are then substituted into (9) to obtain $F(u)$ for $u = 0, 1, 2, \dots, M-1$
- The other half then follows directly from (10) without additional transform evaluations.

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Discussion and Classwork

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Lecture – 08

Harimohan Khatri

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In this lecture...

- i. Smoothing Frequency Domain Filters
- ii. Sharpening Frequency Domain Filters

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Steps for Filtering in Frequency Domain

1. Given an input image $f(x,y)$ of size $M \times N$, obtain the padding sizes P and Q where, $P = 2M$ and $Q = 2N$.
2. Form a padded image $f_p(x,y)$ of size $P \times Q$ using zero-, mirror-, or replicate padding
3. Multiply $f_p(x,y)$ by $(-1)^{x+y}$ to center the Fourier transform on the $P \times Q$ frequency rectangle.
4. Compute the DFT, $F(u,v)$, of the image.
5. Construct a real, symmetric filter transfer function, $H(u,v)$, of size $P \times Q$ with center at $(P/2, Q/2)$.
6. Form the product $G(u,v) = H(u,v)F(u,v)$ using elementwise multiplication i.e $G(i,k) = H(i,k)F(i,k)$ for $i = 0,1,2,3,\dots,M-1$ and $k = 0,1,2,3,\dots,N-1$.

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Steps for Filtering in Frequency Domain

7. Obtain the filtered image by computing the IDFT of $G(u,v)$:

$$g_p(x,y) = (\text{real}[\mathcal{F}^{-1}\{G(u,v)\}])(-1)^{x+y}$$
8. Obtain the final filtered result, $g(x,y)$ by extracting the $M \times N$ region from the top, left quadrant of $g_p(x,y)$.

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Image Smoothing using Lowpass Frequency Domain Filters

- Smoothing is achieved in the frequency domain by dropping out the high frequency components
- The basic model for filtering is:

$$G(u,v) = H(u,v)F(u,v)$$
 where $F(u,v)$ is the Fourier transform of the image being filtered and $H(u,v)$ is the filter transform function
- Low pass filters – only pass the low frequencies, drop the high ones
 - Ideal
 - Butterworth
 - Gaussian

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Ideal Lowpass Filters

- Simply cut off all high frequency components that are a specified distance D_0 from the origin of the transform
- It is specified by the transfer function:

$$H(u,v) = \begin{cases} 1 & \text{if } D(u,v) \leq D_0 \\ 0 & \text{if } D(u,v) > D_0 \end{cases}$$

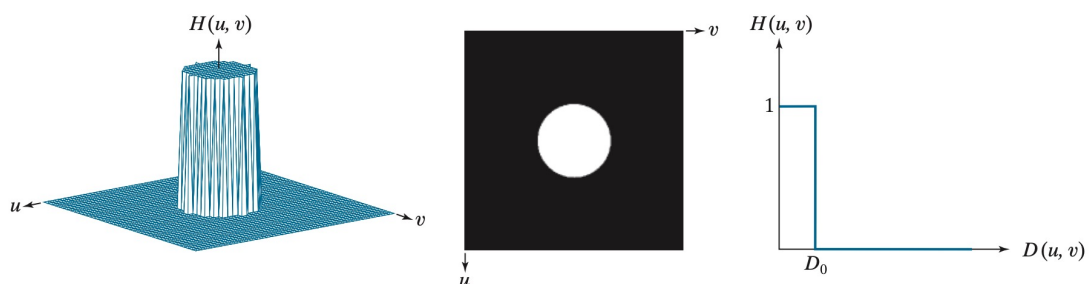
Where D_0 is a positive constant, and $D(u,v)$ is the distance between point (u,v) in the frequency domain and the center of the $P \times Q$ rectangle.

$$D(u,v) = \left[(u - P/2)^2 + (v - Q/2)^2 \right]^{1/2}$$

- changing the distance changes the behavior of the filter

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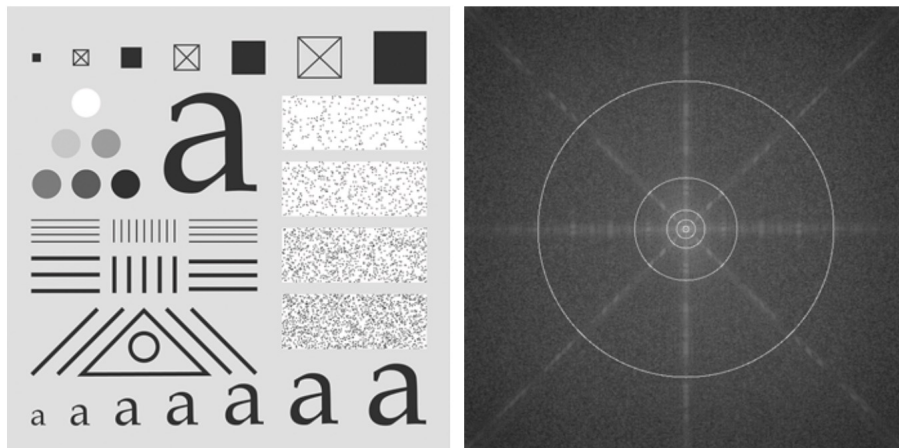
Ideal Lowpass Filter



Transfer function of the Ideal Lowpass Filter

50

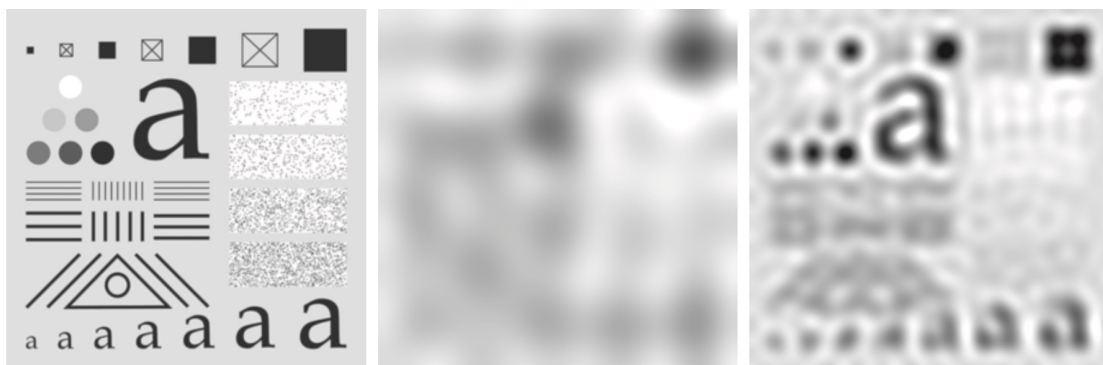
Ideal Lowpass Filter



a) Test pattern of size 688×688 pixels, and (b) its spectrum.

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Ideal Lowpass Filter



(a) Original image of size 688*688 pixels.
(b) and (c) Results of filtering using ILPFs with cutoff frequencies set at radii values 10, 30

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Ideal Lowpass Filter



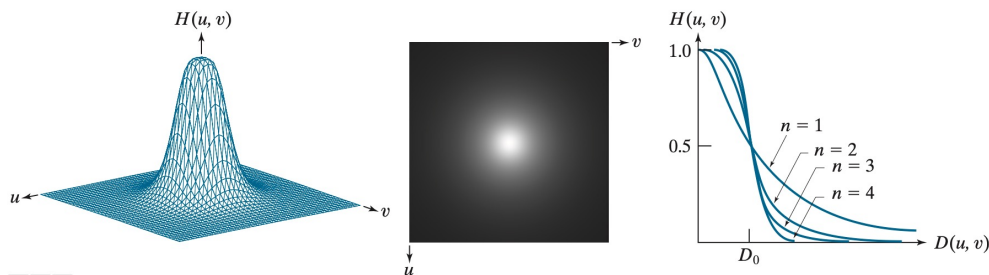
Results of filtering using ILPFs with cutoff frequencies set at radii values 60, 160, and 460,

53

Butterworth Lowpass Filter

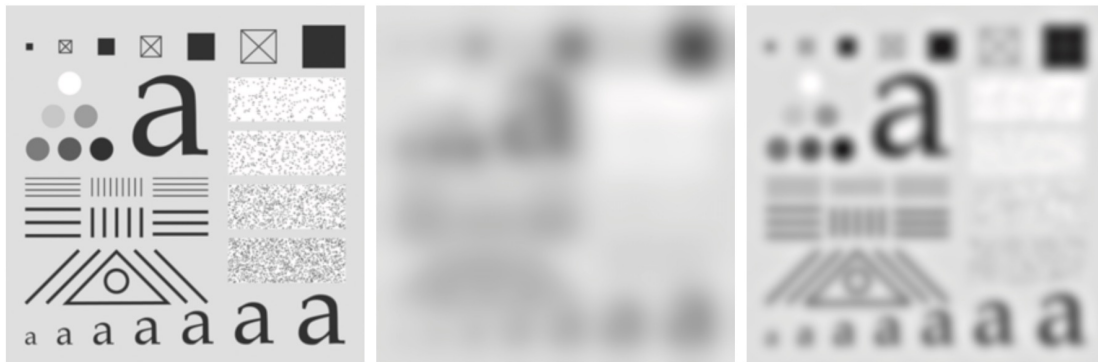
- The transfer function of a Butterworth lowpass filter of order n with cutoff frequency at distance D_0 from the origin is defined as:

$$H(u, v) = \frac{1}{1 + [D(u, v) / D_0]^{2n}}$$



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Butterworth Lowpass Filter



(a) Original image of size 688*688 pixels.

(b) and (c) Results of filtering using BLPFs with cutoff frequencies set at radii values 10, 30 and $n=2.25$

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Butterworth Lowpass Filter



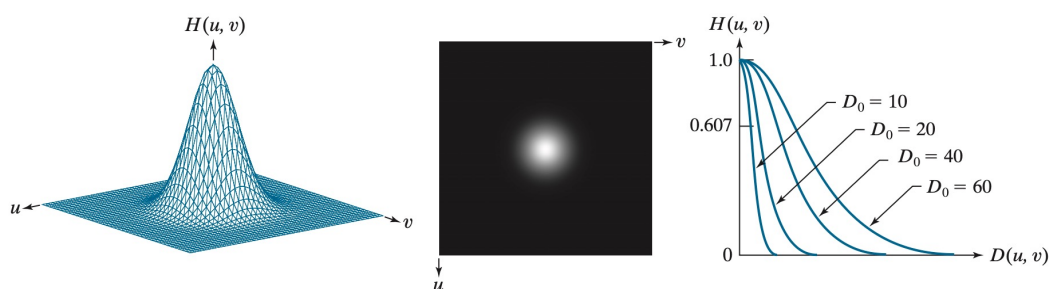
Results of filtering using ILPFs with cutoff frequencies set at radii values 60, 160, and 460 where $n=2.25$

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Gaussian Lowpass Filter

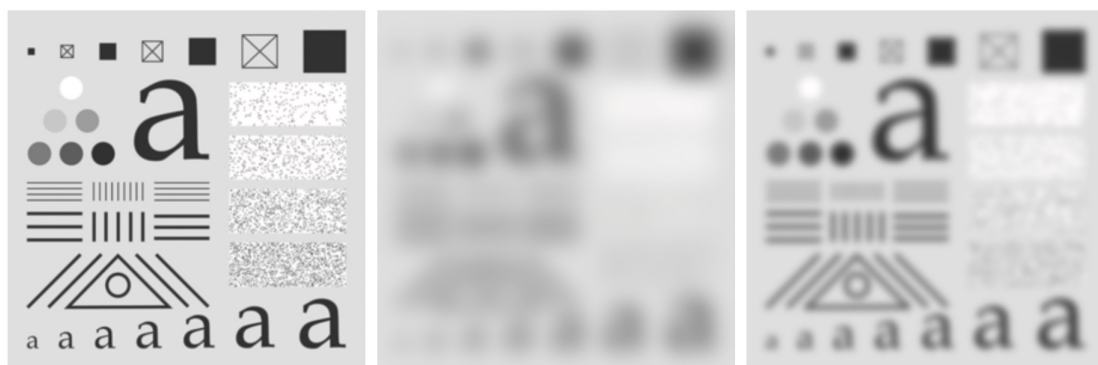
- The transfer function of a Gaussian lowpass filter is defined as:

$$H(u, v) = e^{-D^2(u, v)/2\sigma^2}$$



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Gaussian Lowpass Filter



- (a) Original image of size 688*688 pixels.
 (b) and (c) Results of filtering using GLPFs with cutoff frequencies set at radii values 10, 30

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Gaussian Lowpass Filter



Results of filtering using ILPFs with cutoff frequencies set at radii values 60, 160, and 460

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Lowpass Filtering Compared



Result of filtering with ideal low pass filter of radius 15

Result of filtering with Butterworth filter of order 2 and cutoff radius 15

Result of filtering with Gaussian filter with cutoff radius 15

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Lowpass Filtering Example

- A low pass Gaussian filter is used to connect broken text

Historically, certain computer programs were written using only two digits rather than four to define the applicable year. Accordingly, the company's software may recognize a date using "00" as 1900 rather than the year 2000.

ea

Historically, certain computer programs were written using only two digits rather than four to define the applicable year. Accordingly, the company's software may recognize a date using "00" as 1900 rather than the year 2000.

ea

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Lowpass Filtering Example

- Different lowpass Gaussian filters used to remove blemishes in a photograph



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Image Sharpning using Highpass Filters

- High pass filters – only pass the high frequencies, drop the low ones
- High pass frequencies are precisely the reverse of low pass filters.
- That means, subtracting a lowpass filter transfer function from 1 yields the corresponding highpass filter transfer function in the frequency domain:

$$H_{HP}(u, v) = 1 - H_{LP}(u, v)$$

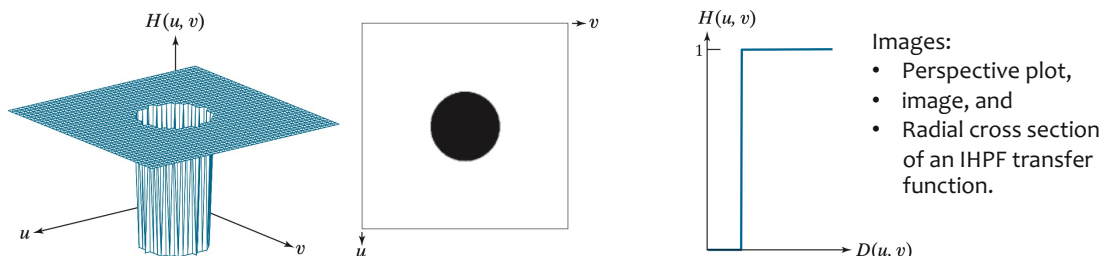
63

Image Sharpning using Idle Highpass Filter

- The ideal high pass filter is given as:

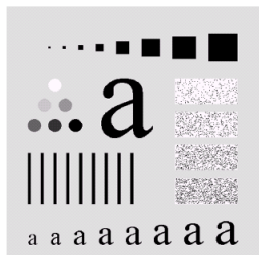
$$H(u, v) = \begin{cases} 0 & \text{if } D(u, v) \leq D_0 \\ 1 & \text{if } D(u, v) > D_0 \end{cases}$$

where D_0 is the cut off distance as before

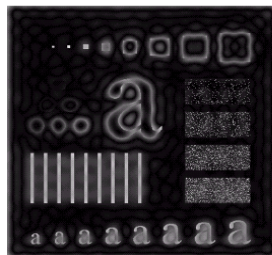


64

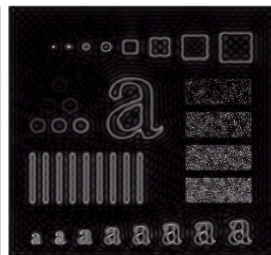
Image Sharpning using Idle Highpass Filter



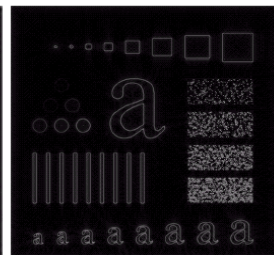
Original image of size
688*688 pixels.



Results of ideal high pass
filtering with $D_0 = 15$



Results of ideal high
pass filtering with
 $D_0 = 30$



Results of ideal high pass
filtering with $D_0 = 80$

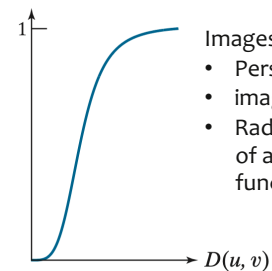
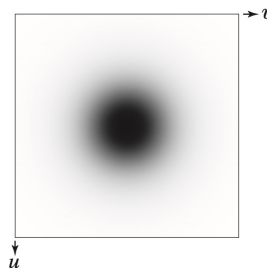
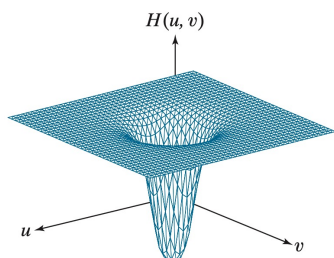
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Image Sharpning using Butterworth Filter

- The Butterworth high pass filter is given as:

$$H(u, v) = \frac{1}{1 + [D_0 / D(u, v)]^{2n}}$$

where n is the order and D_0 is the cut off distance as before

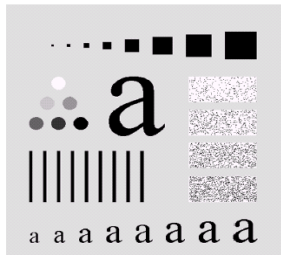


Images:

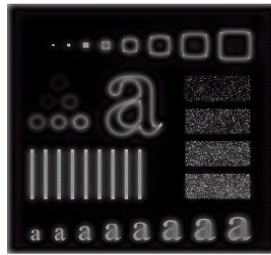
- Perspective plot,
- image, and
- Radial cross section of an BHPF transfer function.

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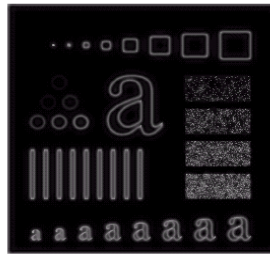
Image Sharpning using Butterworth Filter



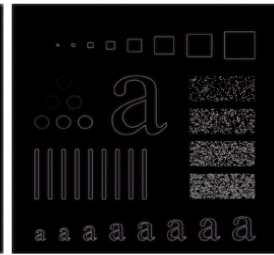
Original image of size
688*688 pixels.



Results of Butterworth
high pass filtering of
order $n=2$ with $D_0=15$



Results of Butterworth
high pass filtering of
order $n=2$ with
 $D_0=30$



Results of Butterworth
high pass filtering of
order $n=2$ with $D_0=80$

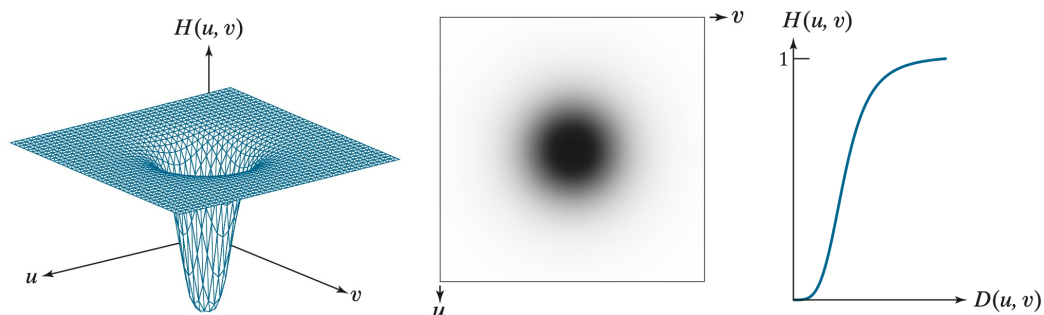
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Image Sharpning using Gaussian Highpass Filter

- The Gaussian high pass filter is given as:

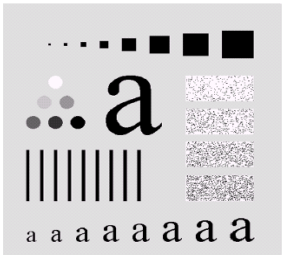
$$H(u, v) = 1 - e^{-D^2(u, v) / 2D_0^2}$$

where D_0 is the cut off distance



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Image Sharpning using Butterworth Filter



Original image of size 688*688 pixels.



Results of Gaussian high pass filtering with $D_0 = 15$



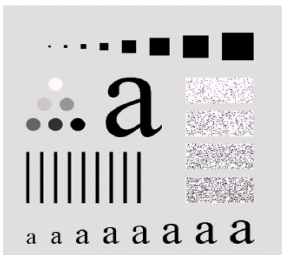
Results of Gaussian high pass filtering with $D_0 = 30$



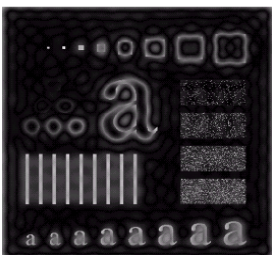
Results of Gaussian high pass filtering with $D_0 = 80$

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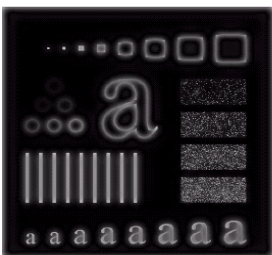
Highpass Filter Comparison



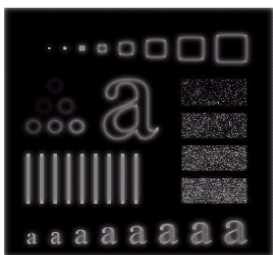
Original image of size 688*688 pixels.



Results of ideal high pass filtering with $D_0 = 15$



Results of Butterworth high pass filtering of order 2 with $D_0 = 15$



Results of Gaussian high pass filtering with $D_0 = 15$

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Laplacian in Frequency Domain

- 2-D Laplacian in spatial domain is defined as:

$$\nabla^2 f(x, y) = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

- The Fourier transform of a derivative is:

$$\mathcal{F}\left\{\frac{\partial^2 f(x, y)}{\partial x^2}\right\} = -(2\pi u)^2 F(u, v)$$

$$\mathcal{F}\left\{\frac{\partial^2 f(x, y)}{\partial y^2}\right\} = -(2\pi v)^2 F(u, v)$$

- Then, Laplacian in frequency domain is:

$$\mathcal{F}\{\nabla^2 f(x, y)\} = \mathcal{F}\left\{\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}\right\} = \mathcal{F}\left\{\frac{\partial^2 f}{\partial x^2}\right\} + \mathcal{F}\left\{\frac{\partial^2 f}{\partial y^2}\right\}$$

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Laplacian in Frequency Domain

- Substituting the derivative transforms:

$$\Rightarrow -(2\pi u)^2 F(u, v) - (2\pi v)^2 F(u, v)$$

$$\Rightarrow -4\pi^2(u^2 + v^2) F(u, v)$$

- To apply laplacian in frequency domain:

1. Compute $F(u, v) = \mathcal{F}\{f(x, y)\}$
2. Multiply by $H(u, v) = -4\pi^2(u^2 + v^2)$
3. Inverse Fourier Transform $\rightarrow$ Result is $\nabla^2 f(x, y)$

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Discussion and Classwork

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Lecture – 09

Harimohan Khatri

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In this lecture...

- i. Hadamard Transform
- ii. Haar Transform
- iii. Discrete Cosine Transform

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Hadamard Transform

- The Hadamard transform is a linear , orthogonal and non-sinusoidal transform
- It has all its coefficient real and their values will be either +1 or -1 only.
- It has vary low computational complexity
- The 1D forward Hadamard Kernel is given by the relation

$$g(x,u) = \frac{1}{\sqrt{N}} (-1)^{\sum_{i=0}^{P-1} bi(x).bi(u)}$$

- In function form,

$$F(x,u) = \frac{1}{\sqrt{N}} \sum_{x=0}^{M-1} f(x) (-1)^{\sum_{i=0}^{P-1} bi(x)bi(u)}$$

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Hadamard Transform

- 2-D Hadamard transform is

$$g(x,y,u,v) = \frac{1}{\sqrt{N}} (-1)^{\sum_{i=0}^{P-1} bi(x).bi(u)+bi(y).bi(v)}$$

- 2-D Function is,

$$F(u,v) = \frac{1}{\sqrt{N}} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x,y) (-1)^{\sum_{i=0}^{P-1} bi(x).bi(u)+bi(y).bi(v)}$$

where, N=order of function

x = time index

u = frequency index

P = no of bits required to represent N

p = $\log_2 N$

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Kernel design of Hadamard Transform of order N=2

Step-1: To determine the kernel for Hadamard Transform, Let use first consider the equation

$$g(x,y) = \frac{1}{\sqrt{N}} (-1)^{\sum_{i=0}^{P-1} bi(x).bi(u)}$$

Step-2: For N=2, the values time and frequency indexes of 'x' and 'u' are 0,1. and size of the matrix is 2×2.

Step-3: No of bits required to represent order, $P = \log_2 N = \log_2(2)=1$

Step-4: Determine the values of $bi(x)$, as the value of N=2, x=0,1 (0→N-1) (Representation in binary form)

for n = 0, $b_0(n) = 0$ and n = 1, $b_0(n)=1$

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Kernel design of Hadamard Transform of order N=2

Step-5: Determine the coefficient for every point of the transfer function $g(x,u)$, by expanding Hadamard function and substituting the binary parameter values obtained in step-4.

$$\begin{aligned} g(x,y) &= \frac{1}{\sqrt{N}} (-1)^{\sum_{i=0}^{P-1} b_i(x).b_i(u)} \\ &= \frac{1}{\sqrt{2}} (-1)^{\sum_{i=0}^0 b_i(x).b_i(u)} \\ &= \frac{1}{\sqrt{2}} (-1)^{b_0(x).b_0(u)} \end{aligned}$$

Step-6: Determine the varies values of $g(x,u)$ by substituting the values of x and u then, we get

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Kernel design of Hadamard Transform of order N=2

$$\begin{aligned} g(0,0) &= \frac{1}{\sqrt{2}} (-1)^{b_0(0).b_0(0)} \\ &= \frac{1}{\sqrt{2}} (-1)^0 \\ &= \frac{1}{\sqrt{2}} \end{aligned}$$

$$\begin{aligned} g(0,1) &= \frac{1}{\sqrt{2}} (-1)^{b_0(0).b_0(1)} \\ &= \frac{1}{\sqrt{2}} (-1)^{0.1} \\ &= \frac{1}{\sqrt{2}} (-1)^0 = \frac{1}{\sqrt{2}} \end{aligned}$$

$$\begin{aligned} g(1,0) &= \frac{1}{\sqrt{2}} (-1)^{b_0(1).b_0(0)} \\ &= \frac{1}{\sqrt{2}} (-1)^{1.0} \\ &= \frac{1}{\sqrt{2}} (-1)^{1.0} = \frac{1}{\sqrt{2}} \end{aligned}$$

$$\begin{aligned} g(1,1) &= \frac{1}{\sqrt{2}} (-1)^{b_0(1).b_0(1)} \\ &= \frac{1}{\sqrt{2}} (-1)^{1.1} \\ &= \frac{1}{\sqrt{2}} (-1)^1 = -\frac{1}{\sqrt{2}} \end{aligned}$$

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Kernel design of Hadamard Transform of order N=2

- Final kernel of 2×2 Hadamard transform is given by;

$$\begin{array}{cc}
 & \begin{array}{cc} x=0 & x=1 \end{array} \\
 \begin{array}{c} u=0 \\ u=1 \end{array} & \begin{array}{|c|c|} \hline 1/\sqrt{2} & 1/\sqrt{2} \\ \hline 1/\sqrt{2} & -1/\sqrt{2} \\ \hline \end{array}
 \end{array}
 \Rightarrow
 \frac{1}{\sqrt{2}}
 \begin{array}{|c|c|} \hline 1 & 1 \\ \hline 1 & -1 \\ \hline \end{array}$$

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Kernel design of Hadamard Transform of order N=4

Step-1: To determine the kernel for Hadamard Transform, Let use first consider the equation

$$g(x,y) = \frac{1}{\sqrt{N}} (-1)^{\sum_{i=0}^{P-1} b_i(x) \cdot b_i(y)}$$

Step-2: Size of kernel, N=4 so x and u varies from 0 to N-1 i.e takes values [0,1,2,3].

Step-3: No of bits required to represent order 4, $P = \log_2 N = \log_2(4)=2$

Step-4: Representing all x = 0,1,2,3 values in binary,

$$n = 0, b_1(n) = 0 \text{ and } b_0(n) = 0$$

$$n = 1, b_1(n) = 0 \text{ and } b_0(n) = 1$$

$$n = 2, b_1(n) = 1 \text{ and } b_0(n) = 0$$

$$n = 3, b_1(n) = 1 \text{ and } b_0(n) = 1$$

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Kernel design of Hadamard Transform of order N=4

Step-5: Determine the coefficient for every point of transfer function $g(x,u)$ by expanding Hadamard function and substituting the binary parameter values obtained in step 4:

$$\begin{aligned} g(x,y) &= \frac{1}{\sqrt{4}} (-1)^{\sum_{i=0}^{2-1} b_i(x).b_i(u)} \\ &= \frac{1}{\sqrt{4}} (-1)^{\sum_{i=0}^1 b_i(x).b_i(u)} \\ &= \frac{1}{\sqrt{4}} (-1)^{b_0(x).b_0(u)+b_1(x).b_1(u)} \end{aligned}$$

Step-6: Determining the varies values of $g(x,u)$ by substituting values of x and u.

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Kernel design of Hadamard Transform of order N=4

$$\begin{aligned} g(0,0) &= \frac{1}{\sqrt{4}} (-1)^{b_0(0).b_0(0)+b_1(0).b_1(0)} & g(0,2) &= \frac{1}{\sqrt{4}} (-1)^{b_0(0).b_0(2)+b_1(0).b_1(2)} \\ &= \frac{1}{\sqrt{4}} (-1)^{b_0(0).b_0(0)+b_1(0).b_1(0)} & &= \frac{1}{\sqrt{4}} (-1)^{0.0+0.1} \\ &= \frac{1}{\sqrt{4}} (-1)^{0.0+0.0} & &= \frac{1}{\sqrt{4}} (-1)^0 = \frac{1}{\sqrt{4}} \\ &= \frac{1}{\sqrt{4}} (-1)^0 = \frac{1}{\sqrt{4}} \\ g(0,1) &= \frac{1}{\sqrt{4}} (-1)^{b_0(0).b_0(1)+b_1(0).b_1(1)} & g(0,3) &= \frac{1}{\sqrt{4}} (-1)^{b_0(0).b_0(3)+b_1(0).b_1(3)} \\ &= \frac{1}{\sqrt{4}} (-1)^{0.1+0.0} & &= \frac{1}{\sqrt{4}} (-1)^{0.1+0.1} \\ &= \frac{1}{\sqrt{4}} (-1)^0 = \frac{1}{\sqrt{4}} & &= \frac{1}{\sqrt{4}} (-1)^0 = \frac{1}{\sqrt{4}} \end{aligned}$$

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Kernel design of Hadamard Transform of order N=4

$$\begin{aligned}
 g(1,0) &= \frac{1}{\sqrt{4}} (-1)^{b_0(x).b_0(u)+b_1(x).b_1(u)} \\
 &= \frac{1}{\sqrt{4}} (-1)^{b_0(1).b_0(0)+b_1(1).b_1(0)} \\
 &= \frac{1}{\sqrt{4}} (-1)^{1.0+0.0} \\
 &= \frac{1}{\sqrt{4}} (-1)^0 = \frac{1}{\sqrt{4}}
 \end{aligned}$$

$$\begin{aligned}
 g(1,1) &= \frac{1}{\sqrt{4}} (-1)^{b_0(1).b_0(1)+b_1(1).b_1(1)} \\
 &= \frac{1}{\sqrt{4}} (-1)^{1.1+0.1} \\
 &= \frac{1}{\sqrt{4}} (-1)^1 = -\frac{1}{\sqrt{4}}
 \end{aligned}$$

$$\begin{aligned}
 g(1,2) &= \frac{1}{\sqrt{4}} (-1)^{b_0(1).b_0(2)+b_1(1).b_1(2)} \\
 &= \frac{1}{\sqrt{4}} (-1)^{1.0+0.1} \\
 &= \frac{1}{\sqrt{4}} (-1)^0 = \frac{1}{\sqrt{4}}
 \end{aligned}$$

$$\begin{aligned}
 g(1,3) &= \frac{1}{\sqrt{4}} (-1)^{b_0(1).b_0(3)+b_1(1).b_1(3)} \\
 &= \frac{1}{\sqrt{4}} (-1)^{1.1+0.1} \\
 &= \frac{1}{\sqrt{4}} (-1)^1 = -\frac{1}{\sqrt{4}}
 \end{aligned}$$

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Kernel design of Hadamard Transform of order N=4

$$\begin{aligned}
 g(2,0) &= \frac{1}{\sqrt{4}} (-1)^{b_0(x).b_0(u)+b_1(x).b_1(u)} \\
 &= \frac{1}{\sqrt{4}} (-1)^{b_0(2).b_0(0)+b_1(2).b_1(0)} \\
 &= \frac{1}{\sqrt{4}} (-1)^{0.0+1.0} \\
 &= \frac{1}{\sqrt{4}} (-1)^0 = \frac{1}{\sqrt{4}}
 \end{aligned}$$

$$\begin{aligned}
 g(2,1) &= \frac{1}{\sqrt{4}} (-1)^{b_0(2).b_0(1)+b_1(2).b_1(1)} \\
 &= \frac{1}{\sqrt{4}} (-1)^{0.1+1.0} \\
 &= \frac{1}{\sqrt{4}} (-1)^0 = \frac{1}{\sqrt{4}}
 \end{aligned}$$

$$\begin{aligned}
 g(2,2) &= \frac{1}{\sqrt{4}} (-1)^{b_0(2).b_0(2)+b_1(2).b_1(2)} \\
 &= \frac{1}{\sqrt{4}} (-1)^{0.0+1.1} \\
 &= \frac{1}{\sqrt{4}} (-1)^1 = -\frac{1}{\sqrt{4}}
 \end{aligned}$$

$$\begin{aligned}
 g(2,3) &= \frac{1}{\sqrt{4}} (-1)^{b_0(2).b_0(3)+b_1(2).b_1(3)} \\
 &= \frac{1}{\sqrt{4}} (-1)^{0.1+1.1} \\
 &= \frac{1}{\sqrt{4}} (-1)^1 = -\frac{1}{\sqrt{4}}
 \end{aligned}$$

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Kernel design of Hadamard Transform of order N=4

$$\begin{aligned} g(3,0) &= \frac{1}{\sqrt{4}} (-1)^{b_0(x).b_0(u)+b_1(x).b_1(u)} \\ &= \frac{1}{\sqrt{4}} (-1)^{b_0(3).b_0(0)+b_1(3).b_1(0)} \\ &= \frac{1}{\sqrt{4}} (-1)^{1.0+1.0} \\ &= \frac{1}{\sqrt{4}} (-1)^0 = \frac{1}{\sqrt{4}} \end{aligned}$$

$$\begin{aligned} g(3,1) &= \frac{1}{\sqrt{4}} (-1)^{b_0(3).b_0(1)+b_1(3).b_1(1)} \\ &= \frac{1}{\sqrt{4}} (-1)^{1.1+1.0} \\ &= \frac{1}{\sqrt{4}} (-1)^1 = -\frac{1}{\sqrt{4}} \end{aligned}$$

$$\begin{aligned} g(3,2) &= \frac{1}{\sqrt{4}} (-1)^{b_0(3).b_0(2)+b_1(3).b_1(2)} \\ &= \frac{1}{\sqrt{4}} (-1)^{1.0+1.1} \\ &= \frac{1}{\sqrt{4}} (-1)^1 = -\frac{1}{\sqrt{4}} \end{aligned}$$

$$\begin{aligned} g(3,3) &= \frac{1}{\sqrt{4}} (-1)^{b_0(3).b_0(3)+b_1(3).b_1(3)} \\ &= \frac{1}{\sqrt{4}} (-1)^{1.1+1.1} \\ &= \frac{1}{\sqrt{4}} (-1)^1 = -\frac{1}{\sqrt{4}} \end{aligned}$$

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Kernel design of Hadamard Transform of order N=2

- Final kernel of 4×4 Hadamard transform is given by;

	x=0	x=1	x=2	x=3
u=0	$1/\sqrt{4}$	$1/\sqrt{4}$	$1/\sqrt{4}$	$1/\sqrt{4}$
u=1	$1/\sqrt{4}$	$-1/\sqrt{4}$	$1/\sqrt{4}$	$-1/\sqrt{4}$
u=2	$1/\sqrt{4}$	$1/\sqrt{4}$	$-1/\sqrt{4}$	$-1/\sqrt{4}$
u=3	$1/\sqrt{4}$	$-1/\sqrt{4}$	$-1/\sqrt{4}$	$1/\sqrt{4}$

 $\Rightarrow \frac{1}{\sqrt{4}} \begin{bmatrix} \boxed{1} & \boxed{1} & \boxed{1} & \boxed{1} \\ \boxed{1} & \boxed{-1} & \boxed{1} & \boxed{-1} \\ \boxed{1} & \boxed{1} & \boxed{-1} & \boxed{-1} \\ \boxed{1} & \boxed{-1} & \boxed{-1} & \boxed{1} \end{bmatrix} \Rightarrow \frac{1}{\sqrt{4}} \begin{bmatrix} H_2 & H_2 \\ H_2 & -H_2 \end{bmatrix}$

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Kernel design of Hadamard Transform of order N=8

- Kernel for N=2 is:

$$K_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

- Kernel for N=4 is:

$$K_4 = \frac{1}{\sqrt{4}} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix}$$

- Generalizing the filters,

$$H_N = \frac{1}{\sqrt{N}} \begin{bmatrix} H_{N/2} & H_{N/2} \\ H_{N/2} & -H_{N/2} \end{bmatrix}$$

- So,

$$H_8 = \frac{1}{\sqrt{8}} \begin{bmatrix} H_4 & H_4 \\ H_4 & -H_4 \end{bmatrix}$$

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Kernel design of Hadamard Transform of order N=8

- Kernel for N=8 is:

$$H_8 = \frac{1}{\sqrt{8}}$$

1	1	1	1	1	1	1	1
1	-1	1	-1	1	-1	1	-1
1	1	-1	-1	1	1	-1	-1
1	-1	-1	1	1	-1	-1	1
1	1	1	1	-1	-1	-1	-1
1	-1	1	-1	-1	1	-1	1
1	1	-1	-1	-1	-1	1	1
1	-1	-1	1	-1	1	1	-1

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Hadamard Matrix Ordering

- Order of the matrix obtained from equation is known as Natural Ordering.
- If we order the matrix on basis of number of sign changed in a row, such ordering is known as sequential ordering.

0	1	1	1	1	1	1	1	1	0
1	1	-1	1	-1	1	-1	1	-1	7
2	1	1	-1	-1	1	1	-1	-1	3
3	1	-1	-1	1	1	-1	-1	1	4
4	1	1	1	1	-1	-1	-1	-1	1
5	1	-1	1	-1	-1	1	-1	1	6
6	1	1	-1	-1	-1	-1	1	1	2
7	1	-1	-1	1	-1	1	1	-1	5

Natural Ordering

No of Sign Changes

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Hadamard Matrix Ordering

Hadamard matrix in sequential ordering:

0	1	1	1	1	1	1	1	1	0
4	1	1	1	1	-1	-1	-1	-1	1
6	1	1	-1	-1	-1	-1	1	1	2
2	1	1	-1	-1	1	1	-1	-1	3
3	1	-1	-1	1	1	-1	-1	1	4
7	1	-1	-1	1	-1	1	1	-1	5
5	1	-1	1	-1	-1	1	-1	1	6
1	1	-1	1	-1	1	-1	1	-1	7

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Hadamard Matrix Ordering

- Bit reverse ordering is obtained by reversing binary bits of natural order

Natural		Bit Reverse	
Decimal	Binary	Bit Reverse	Decimal
0	000	000	0
1	001	100	4
2	010	010	2
3	011	110	6
4	100	001	1
5	101	101	5
6	110	011	3
7	111	111	7

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Hadamard Matrix Ordering

- Bit reverse ordering is obtained by reversing binary bits of natural order

0	1	1	1	1	1	1	1	1
4	1	1	1	1	-1	-1	-1	-1
2	1	1	-1	-1	1	1	-1	-1
6	1	1	-1	-1	-1	-1	1	1
1	1	-1	1	-1	1	-1	1	-1
5	1	-1	1	-1	-1	1	-1	1
3	1	-1	-1	1	1	-1	-1	1
7	1	-1	-1	1	-1	1	1	-1

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Hadamard Matrix Ordering

- Sequential ordering can be achieved by reversing gray level bits.

Natural			Sequential	
Decimal	Binary	Gray	Bit Reverse	Decimal
0	000	000	000	0
1	001	001	100	4
2	010	011	110	6
3	011	010	010	2
4	100	110	011	3
5	101	111	111	7
6	110	101	101	5
7	111	100	001	1

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Haar Transform

- This transform is based on the class of orthogonal, non-sinusoidal, non square functions whose elements are multiples of -1, +1, 0 by the power of $\sqrt{2}$.
- It is a computationally efficient transfer function for N point whose vector requires $2(N-1)$ addition and N multiplications.

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Steps to Calculate Haar Transform Kernels

Step-01: Determine the values of N which represents the order of the system or the size of the kernel

Step-02: Determine the total no of bits required based on the order of the function using the expression $n = \log_2 N$

Step-03: Determine the values of the two specific terms 'p' and 'q'

- i. $p \in [0, n-1]$ i.e $0 \leq p \leq n-1$
- ii. If $p = 0$ then the values of q is either 0 or 1
- iii. If $p \neq 0$ then $q \in [1, 2^p]$ i.e $1 \leq q \leq 2^p$

Step-04: Determine values of k which tells the total no of rows in the kernel by equation

$$K = 2^p + q - 1$$

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Steps to Calculate Haar Transform Kernels

Step-05: Determine the values of z where

$$z \in \left[0, \frac{1}{N}, \frac{2}{N}, \dots, \frac{N-1}{N}\right]$$

Step-06: If $k=0$ then, $H_0(z) = \frac{1}{\sqrt{N}} \forall z$ else,

$$k \neq 0 \text{ then } H_k(z) = H_{pq}(z) = \frac{1}{\sqrt{N}} \begin{cases} +2^{\frac{p}{2}}, & \frac{q-1}{2^p} \leq z < \frac{q-\frac{1}{2}}{2^p} \\ -2^{\frac{p}{2}}, & \frac{q-\frac{1}{2}}{2^p} \leq z < \frac{q}{2^p} \\ 0, & \text{otherwise} \end{cases}$$

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Kernel Design of Haar Transform of Order N=2

Step-01: Here, order N=2 which represents size of kernel

Step-02: Total no of bits required based on order of function

$$n = \log_2(N) = \log_2(2) = 1$$

Step-03: Determine p and q

- p ranges from 0 to n-1 $\rightarrow 0 \leq p \leq 1-1 \rightarrow p=0$
- if p=0 then q can be 0 or 1

Step-04: Determine value of k which is total no of rows in the kernel

$$k = 2^p + q - 1$$

$$\text{For } p=0 \text{ and } q=1 \rightarrow k=2^0+0-1 = 0$$

$$\text{For } p=0 \text{ and } q=1 \rightarrow k=2^0+1-1 = 1$$

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Kernel Design of Haar Transform of Order N=2

Step-05: Value of $z \in \left[0, \frac{1}{N}, \frac{2}{N}, \dots, \frac{N-1}{N}\right]$

$$\text{i.e } 0 \text{ to } \frac{N-1}{N} \rightarrow 0 \text{ to } \frac{2-1}{2} = \frac{1}{2} \rightarrow z \in \left[0 \text{ to } \frac{1}{2}\right]$$

Step-06: For $k = 0$ then $H_0(z) = \frac{1}{\sqrt{N}} \forall z$ which is,

$$H_0(z) = \frac{1}{\sqrt{2}} \forall z \in \left[0 \text{ to } \frac{1}{2}\right]$$

$$\text{For } k \neq 0 \text{ then } H_k(z) = H_{pq}(z) = \frac{1}{\sqrt{N}} \begin{cases} +2^{\frac{p}{2}}, \frac{q-1}{2^p} \leq z < \frac{q-\frac{1}{2}}{2^p} \\ -2^{\frac{p}{2}}, \frac{q-\frac{1}{2}}{2^p} \leq z < \frac{q}{2^p} \\ 0, \text{otherwise} \end{cases}$$

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Kernel Design of Haar Transform of Order N=2

Here, $k=1$, $p=0$ and $q=1$ then

$$\frac{q-1}{2^p} \leq z < \frac{q-\frac{1}{2}}{2^p} \rightarrow \frac{1-1}{2^0} \leq z < \frac{1-\frac{1}{2}}{2^0} \rightarrow 0 \leq z < 0.5$$

$$\text{Also, } \frac{q-\frac{1}{2}}{2^p} \leq z < \frac{q}{2^p} \rightarrow \frac{1-\frac{1}{2}}{2^0} \leq z < \frac{1}{2^0} \rightarrow 0.5 \leq z < 1$$

So, we get the function as,

$$H_{pq}(z) = \frac{1}{\sqrt{N}} \begin{cases} +2^{\frac{p}{2}}; & \text{for } 0 \leq z < 0.5 \\ -2^{\frac{p}{2}}; & \text{for } 0.5 \leq z < 1 \\ 0, & \text{otherwise} \end{cases}$$

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Kernel Design of Haar Transform of Order N=2

- For $z \in \left[0 \text{ to } \frac{1}{2}\right]$

$$H_1(0) = H_{01}(0) = \frac{2^0}{\sqrt{2}} = \frac{1}{\sqrt{2}}$$

$$H_1\left(\frac{1}{2}\right) = H_{01}\left(\frac{1}{2}\right) = -\frac{2^0}{\sqrt{2}} = -\frac{1}{\sqrt{2}}$$

- So, we get the kernel components as,

$$\text{for } K=0, z=0 \quad H_0(z) = \frac{1}{\sqrt{2}}$$

$$\text{for } K=0, z=\frac{1}{2} \quad H_0(z) = \frac{1}{\sqrt{2}}$$

$$\text{for } K=1, z=0 \quad H_1(z) = \frac{1}{\sqrt{2}}$$

$$\text{for } K=1, z=\frac{1}{2} \quad H_1(z) = -\frac{1}{\sqrt{2}}$$

	$z=0$	$z=\frac{1}{2}$
$k=0$	$\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}$
$k=1$	$\frac{1}{\sqrt{2}}$	$-\frac{1}{\sqrt{2}}$

$\frac{1}{\sqrt{2}}$	1	1
	1	-1

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Kernel Design of Haar Transform of Order N=4

Step-01: Here, $N=4$ which represents order of the system or size of the kernel

Step-02: Total no of bits required $n = \log_2(N)$ i.e $n=\log_2(4)=2$

Step-03: Determining 'p' and 'q'

- p ranges from 0 to n-1 i.e $0 \leq p \leq n-1 \rightarrow 0 \leq p \leq 2-1 \rightarrow p=0,1$
- If $p=0$, q will be 0 or 1
- If $p=1$, q will be $1 \leq q \leq 2^p \rightarrow 1 \leq q \leq 2^1 \rightarrow q=1,2$

We get the combination of p and q as, 00, 01, 11, 12

Step-04: Now determining total no of rows in the kernel, i.e

$$k=2^p+q-1$$

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Kernel Design of Haar Transform of Order N=4

If $p=0$ and $q=0 \rightarrow k = 2^0+0-1=0$

If $p=0$ and $q=1 \rightarrow k = 2^0+1-1=1$

If $p=1$ and $q=1 \rightarrow k = 2^1+1-1=2$

If $p=1$ and $q=2 \rightarrow k = 2^1+2-1=3$

Step-05: Based on N, Determine z where $z \in \left[0, \frac{1}{N}, \frac{2}{N}, \dots, \frac{N-1}{N}\right]$

For $N=4$, $z = \left[0, \frac{1}{4}, \frac{2}{4}, \frac{3}{4}\right]$

Step-06: For $k=0$, $H(z) = \frac{1}{\sqrt{N}} \forall z$, i.e $H_0(z) = \frac{1}{\sqrt{4}} = \frac{1}{2}$

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Kernel Design of Haar Transform of Order N=4

For $k=1 \rightarrow p=0, q=1$,

$$H_1(z) = H_{01}(z) = \frac{1}{\sqrt{4}} \begin{cases} +2^{\frac{0}{2}}, \frac{1-1}{2^0} \leq z < \frac{1-\frac{1}{2}}{2^0} \\ -2^{\frac{0}{2}}, \frac{1-\frac{1}{2}}{2^0} \leq z < \frac{1}{2^0} \\ 0, \text{otherwise} \end{cases} = \frac{1}{\sqrt{4}} \begin{cases} +1, 0 \leq z < 0.5 \\ -1, 0.5 \leq z < 1 \\ 0, \text{otherwise} \end{cases}$$

Then we get,

$$H_1(0) = \frac{1}{2} \quad H_1\left(\frac{1}{4}\right) = \frac{1}{2} \quad H_1\left(\frac{2}{4}\right) = -\frac{1}{2} \quad H_1\left(\frac{3}{4}\right) = -\frac{1}{2}$$

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Kernel Design of Haar Transform of Order N=4

For $k=2 \rightarrow p=1, q=1$,

$$H_2(z) = H_{11}(z) = \frac{1}{\sqrt{4}} \begin{cases} +2^{\frac{1}{2}}, \frac{1-1}{2^1} \leq z < \frac{1-\frac{1}{2}}{2^1} \\ -2^{\frac{1}{2}}, \frac{1-\frac{1}{2}}{2^1} \leq z < \frac{1}{2^1} \\ 0, \text{otherwise} \end{cases} = \frac{1}{\sqrt{4}} \begin{cases} +\sqrt{2}, 0 \leq z < 0.25 \\ -\sqrt{2}, 0.25 \leq z < 0.5 \\ 0, \text{otherwise} \end{cases}$$

Then we get,

$$H_2(0) = \frac{\sqrt{2}}{2} = \frac{1}{\sqrt{2}} \quad H_2\left(\frac{1}{4}\right) = -\frac{\sqrt{2}}{2} = -\frac{1}{\sqrt{2}} \quad H_2\left(\frac{2}{4}\right) = 0 \quad H_2\left(\frac{3}{4}\right) = 0$$

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Kernel Design of Haar Transform of Order N=4

For $k=3 \rightarrow p=1, q=2$

$$H_3(z) = H_{12}(z) = \frac{1}{\sqrt{4}} \begin{cases} +2^{\frac{1}{2}}, & \frac{2^{-1}}{2^1} \leq z < \frac{2^{-\frac{1}{2}}}{2^1} \\ -2^{\frac{1}{2}}, & \frac{2^{-\frac{1}{2}}}{2^1} \leq z < \frac{2}{2^1} \\ 0, & \text{otherwise} \end{cases} = \frac{1}{\sqrt{4}} \begin{cases} +\sqrt{2}, & 0.5 \leq z < 0.75 \\ -\sqrt{2}, & 0.75 \leq z < 1 \\ 0, & \text{otherwise} \end{cases}$$

Then we get,

$$H_3(0) = 0 \quad H_2\left(\frac{1}{4}\right) = 0 \quad H_2\left(\frac{2}{4}\right) = \frac{\sqrt{2}}{2} = \frac{1}{\sqrt{2}} \quad H_2\left(\frac{3}{4}\right) = -\frac{\sqrt{2}}{2} = -\frac{1}{\sqrt{2}}$$

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Kernel Design of Haar Transform of Order N=4

Now, we get the Haar transform kernel for N=4 as:

$$H_4 = \begin{matrix} & \begin{matrix} z=0 & z=\frac{1}{4} & z=\frac{2}{4} & z=\frac{3}{4} \end{matrix} \\ \begin{matrix} k=0 \\ k=1 \\ k=2 \\ k=3 \end{matrix} & \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 & 0 \\ 0 & 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} \end{matrix} = \frac{1}{\sqrt{4}} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ \sqrt{2} & -\sqrt{2} & 0 & 0 \\ 0 & 0 & \sqrt{2} & -\sqrt{2} \end{bmatrix}$$

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Kernel Design of Haar Transform of Order N=8

Similarly, we can get the Haar transform kernel for N=8 as:

$$H_8 = \frac{1}{\sqrt{8}} \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 \\ \sqrt{2} & \sqrt{2} & -\sqrt{2} & -\sqrt{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \sqrt{2} & \sqrt{2} & -\sqrt{2} & -\sqrt{2} \\ 2 & -2 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & -2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & -2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & -2 \end{bmatrix}$$

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Discrete Cosine Transform

- The Discrete Cosine Transform is non-symmetric, real and orthogonal.
- The cosine transform has excellent energy compaction for highly correlated data.
- The N×N Discrete Cosine Transform matrix C(u,v) is defined as:

$$C(u,v) = \begin{cases} \frac{1}{\sqrt{N}}, & \text{for } u = 0, 0 \leq v \leq N-1 \\ \sqrt{\frac{2}{N}} \cos \left[\frac{(2v+1)\pi u}{2N} \right], & \text{for } 1 \leq u \leq N-1 \\ & 0 \leq v \leq N-1 \end{cases}$$

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Discrete Cosine Transform

- The 4×4 DCT matrix can be obtained putting the values of u and v in the transform function.

	$v=0$	$v=1$	$v=2$	$v=3$
$u=0$	0.5	0.5	0.5	0.5
$u=1$	0.65	0.27	-0.27	-0.65
$u=2$	0.5	-0.5	-0.5	0.5
$u=3$	0.27	-0.65	0.65	-0.27

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Computing DCT of 1-D Image

- To calculate DCT of 1-D image we need to perform;

$$F = C.f$$

where, F = DCT image, C =DCT matrix and f =Input image

Q) Find DCT of image $f(x) = \{1,3,4,7\}$

0.5	0.5	0.5	0.5	×	1	=	?
0.65	0.27	-0.27	-0.65		3		
0.5	-0.5	-0.5	0.5		4		
0.27	-0.65	0.65	-0.27		7		

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Computing DCT of 2-D Image

- To calculate DCT of 2-D image we need to perform;

$$F = C.f C^T$$

where, F = DCT image, C=DCT matrix and f=Input image

Q) Find DCT of image f(x,y)

0.5	0.5	0.5	0.5
0.65	0.27	-0.27	-0.65
0.5	-0.5	-0.5	0.5
0.27	-0.65	0.65	-0.27

 $\times$

2	5	3	7
5	6	4	3
1	6	3	7
4	7	8	2

 $\times$

0.5	0.65	0.5	0.27
0.5	0.27	-0.5	-0.65
0.5	-0.27	-0.5	0.65
0.5	-0.65	0.5	-0.27

 $=?$

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Computing IDCT of 2-D Image

- To calculate IDCT of 2-D image we need to perform;

$$f = C^T.F. C$$

where, F = DCT image, C=DCT matrix and f=Input image

Q) Find IDCT of image F(u,v)

0.5	0.65	0.5	0.27
0.5	0.27	-0.5	-0.65
0.5	-0.27	-0.5	0.65
0.5	-0.65	0.5	-0.27

 $\times$

 $\times$

0.5	0.5	0.5	0.5
0.65	0.27	-0.27	-0.65
0.5	-0.5	-0.5	0.5
0.27	-0.65	0.65	-0.27

 $=?$

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Discussion and Classwork

- Calculate Hadamard Transform and Inverse Hadamard Transform for given image $f(x,y)$.
- Calculate Haar Transform and Inverse Hadamard Transform for given image $f(x,y)$.

$f(x,y) =$

2	5	3	7
5	6	4	3
1	6	3	7
4	7	8	2